ML-Assisted Hybrid Optimization of a Lower FR3 4Rx4 FBAW Filter

Conventional SLSQP V1.6 versus GPR-Assisted/Powell V0.6 | V0.7 Fine-Grid Verification

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Executive summary

The retained V1.6 conventional SLSQP baseline fails the Q80 and Q60 passband-ripple limits. The V0.6 GPR-assisted hybrid continuation produces a candidate satisfying all specified ripple, bandwidth, and far-rejection constraints on four aligned frequency grids (0.5, 0.25, 0.1, and 0.05 MHz). At the finest grid, its limiting Q80 ripple is 0.699244 dB against 0.700000 dB, leaving 0.000756 dB margin. Grid convergence is strong, but the margin is small; manufacturing and model uncertainty robustness remain unproven.

1. Scope, model and acceptance criteria

The compared designs use identical active RF model sources: vectorized ABCD cascades with 50-ohm ports and MBVD resonators. Three conditions are evaluated: nominal; Q80 with 40 fF per L1 segment; and Q60 with 60 fF per L1 segment. These Q scenarios include both finite-Q resistance and segment parasitics. The nominal condition still includes MBVD resonator losses.

Criterion	Nominal	Q80	Q60
Ripple maximum (dB)	0.50	0.70	0.85
3-dB BW minimum (GHz)	0.75	0.75	0.75
Far rejection minimum (dB)	60	60	60

Worst insertion loss is reported but has no hard acceptance threshold. The bandwidth definition is a contiguous region around the peak sampled passband transmission, relative to that peak minus 3 dB. Far rejection covers the specified 5.5–5.9 and 7.7–8.4 GHz windows, not the entire stopband. Spur and return-loss constraints were not included in this verification.

2. Optimization workflow and attribution

Stage	Algorithm and role	Evidence / limitation
V1.6 baseline	SciPy minimize(method="SLSQP"); 9	Two saved optimization attempts did not converge

	normalized variables; explicit constraints	successfully; selected design retained from V1.2
GPR stage	12 scalar Gaussian-process regressors; upstream candidate generation	121 samples; 97 train / 24 holdout. Candidate-generation and acquisition records not present in reviewed V0.6 package
V0.6 continuation	SciPy minimize(method="Powell") on previously verified V0.5 candidate seeds	No GPR training or prediction occurs inside robust_margin_v06.py
V0.7 verification	Fixed-parameter physics evaluation only	No SLSQP, Powell, retraining or candidate search

Accordingly, "GPR-assisted hybrid optimization" describes the broader research workflow, not the optimizer invoked directly by V0.6. The incomplete upstream candidate provenance limits claims about the independent contribution of GPR.

3. Common-grid V0.7 final results

All values below are from the reported 0.05 MHz endpoint-aligned 3–11 GHz evaluation; values are transcribed from the supplied Codex verification summary rather than independently rerun here.

Design	Condition	Ripple dB	Worst IL dB	BW sampled GHz	BW interp. GHz	Far RJ dB
V1.6	Nominal	0.498534	0.966147	0.817700	0.817732	64.427293
V1.6	Q80	0.712867	1.751843	0.791250	0.791323	64.390834
V1.6	Q60	0.860680	2.068360	0.784700	0.784778	64.332501
V0.6	Nominal	0.494399	0.990285	0.817850	0.817908	64.550938
V0.6	Q80	0.699244	1.772393	0.792400	0.792428	64.511432
V0.6	Q60	0.842217	2.084790	0.785700	0.785752	64.451668

4. Changes and signed margins

Condition	Ripple reduction (dB)	IL increase (dB)	V0.6 ripple margin (dB)
Nominal	0.004135	0.024138	0.005601
Q80	0.013623	0.020550	0.000756
Q60	0.018463	0.016430	0.007783

At 0.05 MHz, V1.6 fails Q80 ripple by 0.012867 dB and Q60 ripple by 0.010680 dB. V0.6 passes every defined criterion. Its minimum sampled BW margin is +0.035700 GHz (35.7 MHz), and its minimum far-rejection margin is +4.451668 dB. Small IL increases are an accepted performance tradeoff, not an improvement.

5. Frequency-grid convergence

For each of two fixed designs, three conditions were evaluated at 0.5, 0.25, 0.1 and 0.05 MHz: 24 design-condition evaluations total and 1,728,024 frequency-point evaluations. No pass/fail changes occurred across the four grids. The largest ripple change from 0.1 to 0.05 MHz was 2.14×10^{-7} dB.

Endpoint inclusion accounts for material differences between some historical results and the newly aligned-grid values.

Reported execution time was 0.200 s, with peak process memory 76.06 MiB. These are observed values reported by Codex for this local run, not general hardware benchmarks. The full per-grid metrics and extrema are held in the local V0.7 CSV artifacts and were not supplied to this report author.

6. Limitations and reproducibility

- Q80 margin is only 0.000756 dB. Numerical grid convergence does not demonstrate tolerance, temperature, fabrication or model-parameter robustness.
- V1.6 selected design was retained from V1.2 after unsuccessful SLSQP attempts; this comparison should not be described as a successful SLSQP optimum.
- Upstream GPR acquisition/ranking code, fitted model artifacts and holdout statistics were absent from the reviewed package; no causal superiority claim for GPR alone is established.
- The local V0.7 manifest reportedly confirms unchanged frozen source/input hashes, but the manifest and detailed CSVs were not attached here for independent verification.
- The acceptance definition does not cover spur checks, return loss, component tolerances, or complete out-of-band rejection.

7. Conclusion and checkpoint decision

Freeze V0.6 as the current fine-grid-feasible ML-assisted hybrid result, and archive V0.7 as its independent numerical verification checkpoint. Preserve the V1.6 conventional baseline and report its Q80/Q60 failures on the same grid. Do not claim manufacturing robustness or a globally optimal design. The next substantive research step, if needed, is controlled parameter-tolerance analysis rather than additional unconstrained optimizer iterations.

Local provenance references

- V1_6_SLSQP/V1_6_report.json; V1_6_SLSQP/run_v16.py
- V0_6_ML_Assisted/Lower_FR3_4Rx4_GPR_V0_6/verified_candidates_v06.csv; robust_margin_v06.py; train_v03.py
- V0_7_Verification/REPORT.md; metrics_and_margins.csv; grid_convergence.csv; passband_extrema.csv; manifest.json